

## ■■ Installation & Setup Guide

Step-by-step instructions to program and run the project

### Prerequisites

- Computer with Windows / macOS / Linux
- 2x ESP32 development boards (one for Transmitter, one for Receiver)
- Micro-USB cables for programming
- All hardware components assembled and wired correctly
- Arduino IDE 1.8.x or 2.x installed
- Internet connection for library downloads

### Step-by-Step Setup

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| Step 1 | <p>Install Arduino IDE</p> <p>Download from: <a href="https://www.arduino.cc/en/software">https://www.arduino.cc/en/software</a></p> <p>Install and launch the application on your computer.</p>                                                                                                                                                                                                                          |
| Step 2 | <p>Add ESP32 Board Support</p> <p>File &gt; Preferences &gt; Additional Board Manager URLs, paste:<br/><a href="https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json">https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json</a></p> <p>Then: Tools &gt; Board &gt; Board Manager → search 'esp32' → Install 'esp32 by Espressif Systems'</p> |
| Step 3 | <p>Install Required Libraries</p> <p>Tools &gt; Manage Libraries, install each:</p> <ul style="list-style-type: none"><li>• MAX30105 by SparkFun (Pulse Oximeter)</li><li>• OneWire by Paul Stoffregen (Temperature sensor)</li><li>• DallasTemperature by Miles Burton</li><li>• LiquidCrystal_I2C by Frank de Brabander</li><li>• ThingSpeak by MathWorks (Cloud dashboard)</li></ul>                                   |
| Step 4 | <p>Clone Repository</p> <p>Open terminal/command prompt:</p> <pre>git clone https://github.com/YOUR_USERNAME/scuba-lifi-health-monitor.git cd scuba-lifi-health-monitor</pre>                                                                                                                                                                                                                                             |
| Step 5 | <p>Upload Transmitter Code</p> <p>Connect Transmitter ESP32 to computer via USB.</p> <p>Open: <code>src/transmitter/transmitter.ino</code></p> <p>Tools &gt; Board &gt; ESP32 Dev Module</p> <p>Tools &gt; Port &gt; Select COM port</p> <p>Click Upload (Ctrl+U)</p>                                                                                                                                                     |

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| <b>Step 6</b> | <p>Upload Receiver Code</p> <p>Connect Receiver ESP32 to computer via USB.</p> <p>Open: src/receiver/receiver.ino</p> <p>Enter your Wi-Fi SSID, Password, and ThingSpeak API Key in the code.</p> <p>Click Upload (Ctrl+U)</p>                                                                                                                                                                                       |
| <b>Step 7</b> | <p>Test the System</p> <p>Open Serial Monitor (baud: 115200) on both ESP32s.</p> <p>Transmitter should print: Heart Rate, SpO2, Temperature values.</p> <p>Receiver should print: Received BPM, SpO2, Temp.</p> <p>Place finger on MAX30102 sensor — values should appear.</p> <p>Shine a flashlight on Li-Fi receiver to simulate transmission.</p> <p>Press panic switch — buzzer should activate immediately.</p> |
| <b>Step 8</b> | <p>View Cloud Dashboard</p> <p>Open <a href="https://thingspeak.com">https://thingspeak.com</a> and log in.</p> <p>Navigate to your channel to view live graphs:</p> <p>Field 1: Heart Rate   Field 2: Temperature   Field 3: SpO2</p> <p>Share channel URL for team monitoring.</p>                                                                                                                                 |

### Troubleshooting

| Problem                  | Solution                                                                               |
|--------------------------|----------------------------------------------------------------------------------------|
| ESP32 port not found     | Install CP2102 or CH340 USB driver. Try different USB cable.                           |
| Upload fails             | Hold BOOT button while clicking Upload; release after upload starts.                   |
| MAX30102 shows 0 BPM     | Ensure firm finger placement. Check I2C address (0x57). Add 4.7kΩ pull-ups on SDA/SCL. |
| Li-Fi no signal received | Align LED and photodiode directly. Reduce ambient light. Check LDR biasing resistors.  |
| LCD shows garbage        | Check I2C address (0x27 or 0x3F). Adjust contrast potentiometer.                       |
| ThingSpeak not updating  | Verify Wi-Fi credentials and API Key. Check internet connectivity.                     |